

Close

YASH YADAV (69)

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```
!pip install -q ultralytics
```

```

 46.6/46.6 kB 2.1 MB/s eta 0:
 1.4/1.4 MB 22.5 MB/s eta 0:00:
 53.2/53.2 kB 4.0 MB/s eta 0:00
 77.7/77.7 kB 6.6 MB/s eta 0:00

```

```

import cv2
import numpy as np
import matplotlib.pyplot as plt

from ultralytics import YOLO
from google.colab import files

print("Libraries imported successfully!")

```

Creating new Ultralytics Settings v0.0.8 file   
 View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.yaml'  
 Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs\_dir=/'  
 Libraries imported successfully!

```

model = YOLO("yolov8n.pt")

print("YOLOv8 model loaded successfully!")

```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt>  
 YOLOv8 model loaded successfully!

```

uploaded = files.upload()

image_path = list(uploaded.keys())[0]

print("Uploaded image:", image_path)

```

pexels-thir...020617.jpg  
**pexels-thirdman-8020617.jpg**(image/jpeg) - 2096444 bytes, last modified: 9/16/2026 - 100% done  
 Saving pexels-thirdman-8020617.jpg to pexels-thirdman-8020617.jpg  
 Uploaded image: pexels-thirdman-8020617.jpg

```

image = cv2.imread(image_path)

image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(10, 7))

```

```
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image



```
results = model(image_path)

print("Object detection completed!")
```

```
image 1/1 /content/pexels-thirdman-8020617.jpg: 640x448 7 persons, 396.9ms
Speed: 26.1ms preprocess, 396.9ms inference, 41.1ms postprocess per image
Object detection completed!
```

```
result_image = results[0].plot()

result_image_rgb = cv2.cvtColor(
    result_image,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(12, 8))
```

```
plt.imshow(result_image_rgb)
plt.title("YOLOv8 Object Detection")
plt.axis("off")
plt.show()
```

### YOLOv8 Object Detection



```
result = results[0]

print("Detected Objects:")
print("-" * 40)

for box in result.boxes:

    class_id = int(box.cls[0])
    confidence = float(box.conf[0])

    class_name = model.names[class_id]

    print(
        f"Object: {class_name} | "
```

```
f"Confidence: {confidence:.2f}"
)
```

Detected Objects:

```
-----
Object: person | Confidence: 0.91
Object: person | Confidence: 0.88
Object: person | Confidence: 0.86
Object: person | Confidence: 0.85
Object: person | Confidence: 0.81
Object: person | Confidence: 0.77
Object: person | Confidence: 0.71
```

```
result = results[0]

print("Detected Objects:")
print("-" * 40)

for box in result.bboxes:

    class_id = int(box.cls[0])
    confidence = float(box.conf[0])

    class_name = model.names[class_id]

    print(
        f"Object: {class_name} | "
        f"Confidence: {confidence:.2f}"
    )
```

Detected Objects:

```
-----
Object: person | Confidence: 0.91
Object: person | Confidence: 0.88
Object: person | Confidence: 0.86
Object: person | Confidence: 0.85
Object: person | Confidence: 0.81
Object: person | Confidence: 0.77
Object: person | Confidence: 0.71
```

```
detected_objects = []

for box in result.bboxes:
    class_id = int(box.cls[0])
    class_name = model.names[class_id]
    detected_objects.append(class_name)

print("Total objects detected:", len(detected_objects))

print("\nObject list:")
for obj in detected_objects:
    print("-", obj)
```

Total objects detected: 7

Object list:

```
- person
- person
- person
- person
- person
- person
- person
```

```
from collections import Counter

object_counts = Counter(detected_objects)

print("Object Counts")
print("-" * 30)

for obj, count in object_counts.items():
    print(f"{obj}: {count}")
```

```
Object Counts
-----
person: 7
```

```
print("Bounding Box Information")
print("-" * 50)

for box in result.bboxes:

    class_id = int(box.cls[0])
    class_name = model.names[class_id]
    confidence = float(box.conf[0])

    x1, y1, x2, y2 = box.xyxy[0].tolist()

    print(f"\nObject: {class_name}")
    print(f"Confidence: {confidence:.2f}")
    print(f"Bounding Box: ({x1:.1f}, {y1:.1f}, {x2:.1f}, {y2:.1f})")
```

```
Bounding Box Information
-----

Object: person
Confidence: 0.91
Bounding Box: (3166.8, 3952.2, 3836.7, 5803.5)

Object: person
Confidence: 0.88
Bounding Box: (1318.4, 4155.6, 1834.7, 5860.3)

Object: person
Confidence: 0.86
Bounding Box: (1702.3, 4008.0, 2278.7, 5846.7)

Object: person
Confidence: 0.85
Bounding Box: (2187.4, 4110.8, 2742.0, 5837.5)
```

```
Object: person
Confidence: 0.81
Bounding Box: (918.0, 3991.5, 1386.7, 5861.4)

Object: person
Confidence: 0.77
Bounding Box: (406.6, 3983.0, 1021.8, 5903.6)

Object: person
Confidence: 0.71
Bounding Box: (2706.1, 4027.9, 3221.8, 5818.5)
```

```
print("=" * 50)
print("OBJECT DETECTION SUMMARY")
print("=" * 50)

print("Model: YOLOv8n")
print("Dataset: COCO pretrained model")
print("Total objects detected:", len(detected_objects))

print("\nDetected Classes:")

for obj, count in object_counts.items():
    print(f"{obj}: {count}")
```

```
=====
OBJECT DETECTION SUMMARY
=====
Model: YOLOv8n
Dataset: COCO pretrained model
Total objects detected: 7

Detected Classes:
person: 7
```

```
results_50 = model(
    image_path,
    conf=0.50
)

filtered_image = results_50[0].plot()

filtered_image_rgb = cv2.cvtColor(
    filtered_image,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(12, 8))
plt.imshow(filtered_image_rgb)
plt.title("YOLOv8 Detection - Confidence > 50%")
plt.axis("off")
plt.show()
```

image 1/1 /content/pexels-thirdman-8020617.jpg: 640x448 7 persons, 131.4ms  
Speed: 5.2ms preprocess, 131.4ms inference, 1.3ms postprocess per image at

### YOLOv8 Detection - Confidence > 50%



```
output_path = "yolov8_detection_result.jpg"

cv2.imwrite(
    output_path,
    filtered_image
)

print("Result saved as:", output_path)
```

Result saved as: yolov8\_detection\_result.jpg

